

RETAIL SALES BUSINESS DATABASE & INSIGHTS REPORT

*Design and Develop a Structured Business Database and Generate Insights to Support
Decision-Making*

Data Science Internship

March 2026

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Duration - 2 Weeks

Github Link: <https://github.com/nisansalasandu/retail-sales-business-database>

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1. Project Overview

This project was completed as one of Task of a Data Science Internship. The goal was to design and develop a structured relational business database using a real-world retail sales dataset, then extract meaningful business insights through SQL queries, trend analysis, and data visualizations.

1.1 Objectives

- ❖ Collect and organize a sample business dataset from Kaggle
- ❖ Design a relational database schema with tables, fields, and relationships
- ❖ Clean and preprocess the dataset using Python and Pandas in Jupyter Notebook
- ❖ Store the cleaned data in Microsoft SQL Server
- ❖ Perform SQL queries to extract meaningful business insights
- ❖ Identify trends, patterns, and gaps in the data
- ❖ Create data summaries to support business decisions
- ❖ Document the database schema, queries, and findings

1.2 Dataset

Source: Kaggle — "Retail Sales Dataset" by Buhari Shehu

URL: <https://www.kaggle.com/datasets/buharishehu/retail-sales-dataset>

Sheet	Rows	Columns	Description
Customers	200	7	Customer demographics — CustomerID, FirstName, LastName, Gender, BirthDate, City, JoinDate
Products	50	6	Product catalog — ProductID, ProductName, Category, SubCategory, UnitPrice, CostPrice
Stores	5	4	Branch locations — StoreID, StoreName, City, Region
Transactions	5,000	8	Sales records — TransactionID, Date, CustomerID, ProductID, StoreID, Quantity, Discount, PaymentMethod

1.3 Tools & Technologies

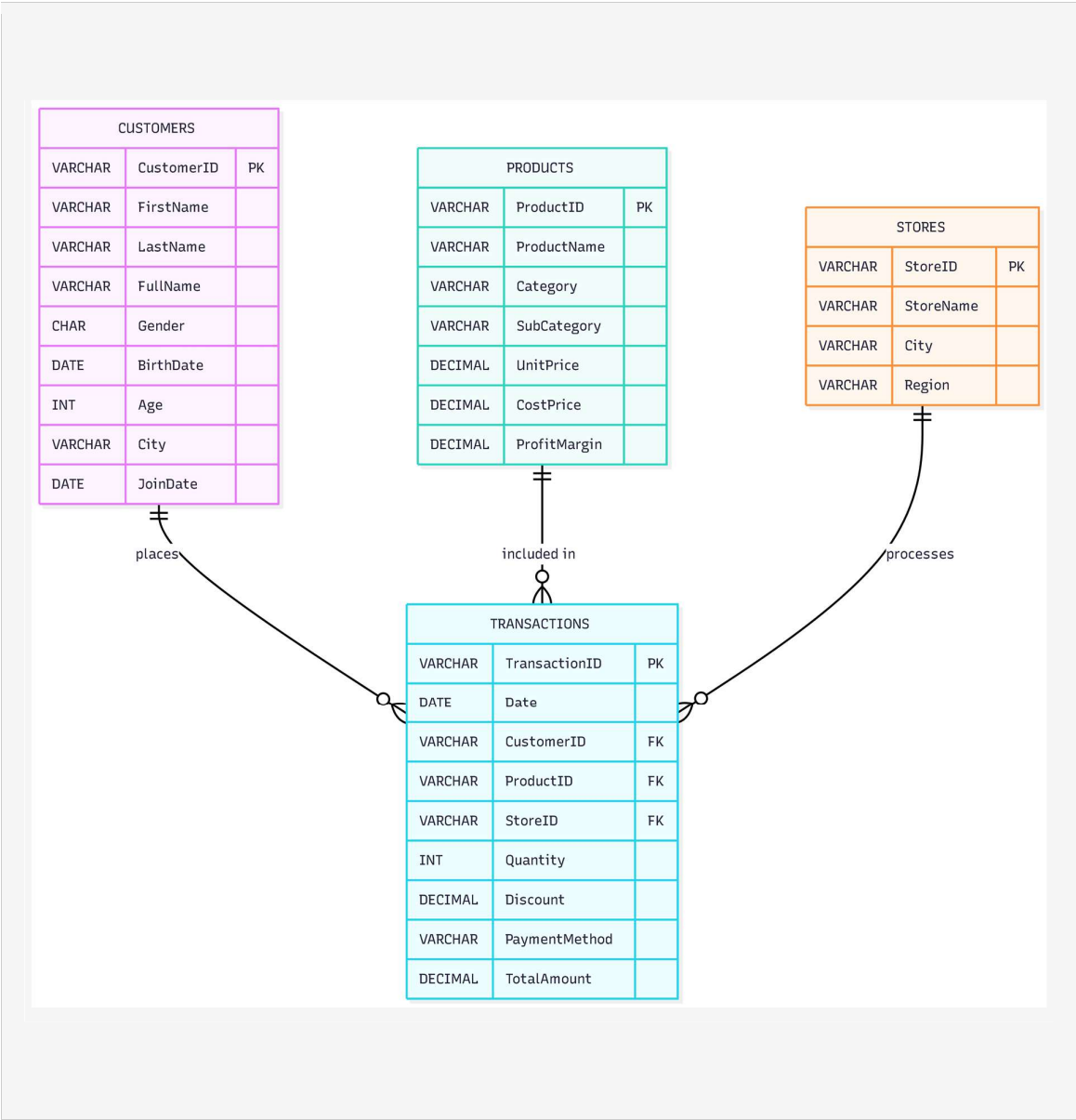
Tool	Purpose
Python 3 / Jupyter Notebook	Data cleaning, preprocessing, EDA, visualizations
Pandas & NumPy	Data manipulation and computation
Matplotlib & Seaborn	Charts and graphical analysis
Microsoft SQL Server	Relational database storage and querying
SQL Server Management Studio (SSMS)	Query execution and database management

2. Database Design

2.1 Relational Schema

The database RetailSalesDataBase follows a star schema pattern, with Transactions as the central fact table and Customers, Products, and Stores as dimension tables. Each dimension table connects to the fact table through foreign key relationships.

2.2 ER Diagram



2.3 Table Definitions

Customers Table

Column	Data Type	Constraint	Description
CustomerID	VARCHAR(10)	PRIMARY KEY	Unique identifier e.g. C001
FirstName	VARCHAR(50)	NOT NULL	Customer first name
LastName	VARCHAR(50)	NOT NULL	Customer last name
FullName	VARCHAR(100)	NOT NULL	Derived: FirstName + LastName
Gender	CHAR(1)	CHECK (M or F)	M = Male, F = Female
BirthDate	DATE	NOT NULL	Date of birth
Age	INT	CHECK (16–100)	Calculated from BirthDate
City	VARCHAR(100)	NOT NULL	Customer city
JoinDate	DATE	NOT NULL	Registration date

Products Table

Column	Data Type	Constraint	Description
ProductID	VARCHAR(10)	PRIMARY KEY	Unique identifier e.g. P001
ProductName	VARCHAR(100)	NOT NULL	Product display name
Category	VARCHAR(50)	NOT NULL	Electronics, Fashion, Groceries
SubCategory	VARCHAR(50)	NOT NULL	Camera, Footwear, Fruits, etc.
UnitPrice	DECIMAL(10,2)	CHECK > 0	Selling price to customer
CostPrice	DECIMAL(10,2)	CHECK > 0	Purchase price from supplier
ProfitMargin	DECIMAL(5,2)	CHECK >= 0	$(\text{UnitPrice} - \text{CostPrice}) / \text{UnitPrice} \times 100$

Stores Table

Column	Data Type	Constraint	Description
StoreID	VARCHAR(10)	PRIMARY KEY	Unique identifier e.g. S001
StoreName	VARCHAR(100)	NOT NULL	Full store name
City	VARCHAR(100)	NOT NULL	Store city location
Region	VARCHAR(50)	NOT NULL	North, South, East, West

Transactions Table (Fact Table)

Column	Data Type	Constraint	Description
TransactionID	VARCHAR(10)	PRIMARY KEY	Unique identifier e.g. T00001
Date	DATE	NOT NULL	Date of transaction
CustomerID	VARCHAR(10)	FK - Customers	References Customers table
ProductID	VARCHAR(10)	FK - Products	References Products table
StoreID	VARCHAR(10)	FK - Stores	References Stores table
Quantity	INT	CHECK > 0	Number of items purchased
Discount	DECIMAL(4,2)	CHECK 0–1	0.00 to 0.15 (0% to 15%)
PaymentMethod	VARCHAR(30)	NOT NULL	Cash / Credit Card / Mobile Money / Bank Transfer
TotalAmount	DECIMAL(12,2)	CHECK > 0	$\text{UnitPrice} \times \text{Quantity} \times (1 - \text{Discount})$

3. Data Cleaning & Preprocessing

Data cleaning was performed in Jupyter Notebook using Python and Pandas (notebook: retail_sales_preprocessing.ipynb). All 4 tables were inspected and cleaned before being exported as CSV files for import into SQL Server.

3.1 Initial Data Quality Assessment

Table	Rows	Missing Values	Duplicates	FK Integrity	Status
Customers	200	0	0	N/A	Clean
Products	50	0	0	N/A	Clean
Stores	5	0	0	N/A	Clean
Transactions	5,000	0	0	Valid	Clean

3.2 Cleaning Actions Performed

Customers

- ❖ Whitespace stripped from all string columns
- ❖ Gender standardized to single uppercase character (M/F)
- ❖ FirstName and LastName title-cased
- ❖ FullName derived column created (FirstName + LastName)
- ❖ Age calculated from BirthDate as of today
- ❖ Sanity check: flagged 5 customers who joined before age 16 (no removal — kept for review)
- ❖ Dates validated and formatted as YYYY-MM-DD

Products

- ❖ Whitespace stripped from all string columns
- ❖ ProductName, Category, SubCategory title-cased
- ❖ Anomaly check: 0 products found where CostPrice $\geq$ UnitPrice
- ❖ ProfitMargin % derived column added: $(\text{UnitPrice} - \text{CostPrice}) / \text{UnitPrice} \times 100$
- ❖ Prices rounded to 2 decimal places

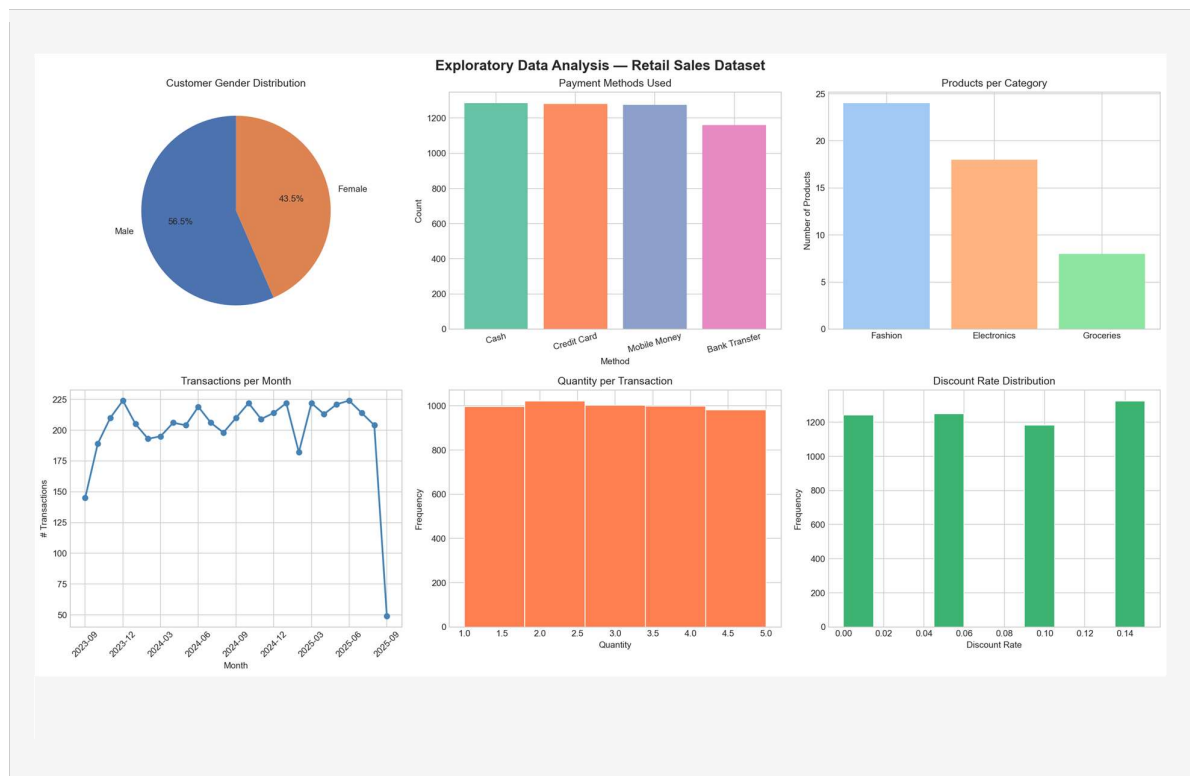
Stores

- ❖ Whitespace stripped from all string columns
- ❖ StoreName, City, Region title-cased
- ❖ Region values validated against expected set (North, South, East, West, Central)

Transactions

- ❖ Whitespace stripped from all string columns
- ❖ Date column validated as datetime
- ❖ Quantity validated: **0 rows found with Quantity <= 0**
- ❖ Discount validated: **0 rows found with Discount outside 0–1 range**
- ❖ PaymentMethod text standardized to title case
- ❖ TotalAmount derived column calculated: $\text{UnitPrice} \times \text{Quantity} \times (1 - \text{Discount})$
- ❖ Duplicate TransactionIDs checked: **0 found**

3.3 Exploratory Data Analysis



3.4 Cleaned Dataset Summary

Table	Rows	Columns (after cleaning)	New Derived Columns
Customers	200	9	FullName, Age
Products	50	7	ProfitMargin
Stores	5	4	None
Transactions	5,000	9	TotalAmount

Business Key Metrics from cleaned data:

Metric	Value
Total Revenue	\$14,301,903.57
Avg Transaction	\$2,860.38
Max Transaction	\$9,763.25
Avg Profit Margin	33.6%
Product Categories	3 (Electronics, Fashion, Groceries)
Date Range	Sep 10, 2023 → Sep 9, 2025

4. Database Setup (Microsoft SQL Server)

4.1 Setup Steps

1. Opened **SQL Server Management Studio (SSMS)** and connected to localhost
2. Created the database: **CREATE DATABASE RetailSalesDataBase;**
3. Ran **01_create_tables.sql** to create all 4 tables with constraints
4. Ran **02_import_data.sql** with BULK INSERT to load all cleaned CSVs
5. Verified row counts and referential integrity post-import

4.2 Row Count Verification

Table	Expected	Imported	Status
Customers	200	200	Correct
Products	50	50	Correct
Stores	5	5	Correct
Transactions	5,000	5,000	Correct

4.3 SQL Views Created

View Name	Description
vw_TransactionDetails	Full joined view of all 4 tables — ready for Power BI or reporting
vw_MonthlySummary	Monthly aggregated sales summary with revenue, transactions, unique customers
vw_CustomerSummary	RFM base per customer — recency, frequency, monetary, first/last purchase

5. Query Results & Business Insights

5.1 Overall Business KPIs

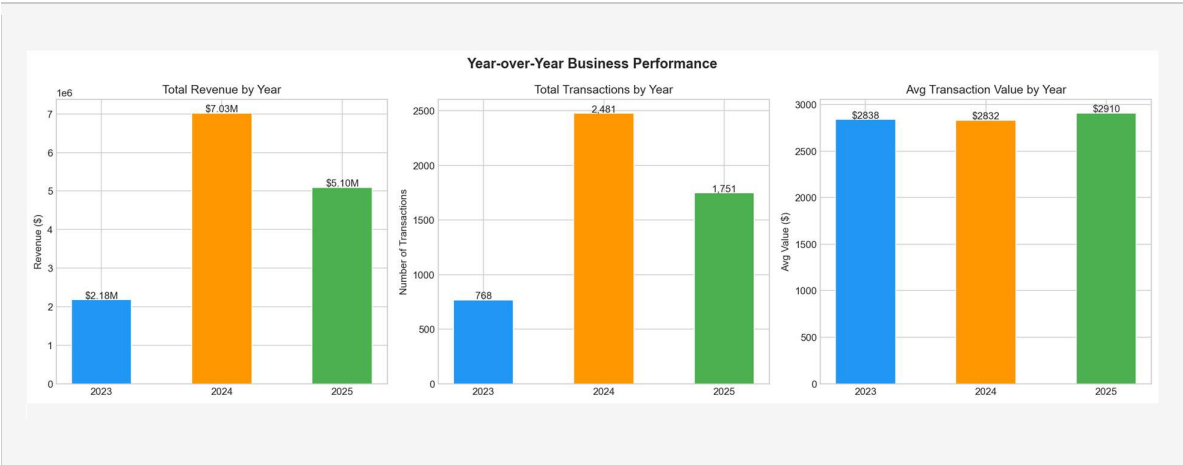
Metric	Value
Total Transactions	5,000
Unique Customers	200
Unique Products Sold	50
Stores Active	5
Total Items Sold	14,949
Total Revenue	\$14,301,903.57
Avg Transaction Value	\$2,860.38
Highest Single Sale	\$9,763.25
Lowest Single Sale	\$21.73
Avg Profit Margin	33.6%
Date Range	Sep 2023 – Sep 2025 (25 months)

5.2 Revenue Performance

Monthly Revenue Trend



Year	Total Revenue	Transactions	Avg Transaction Value
2023 (partial)	\$2,179,931	768	\$2,838
2024 (full)	\$7,026,767	2,481	\$2,832
2025 (partial)	\$5,095,204	1,751	\$2,910



Best & Worst Performing Months

Rank	Month	Revenue	Transactions
#1 Best	2025-04 (April 2025)	\$708,233	213
#2 Best	2025-06 (June 2025)	\$679,817	224
#3 Best	2024-12 (December 2024)	\$663,271	214
Worst	2023-09 (Sep 202)	\$380,992	145

5.3 Product Performance

Revenue by Category

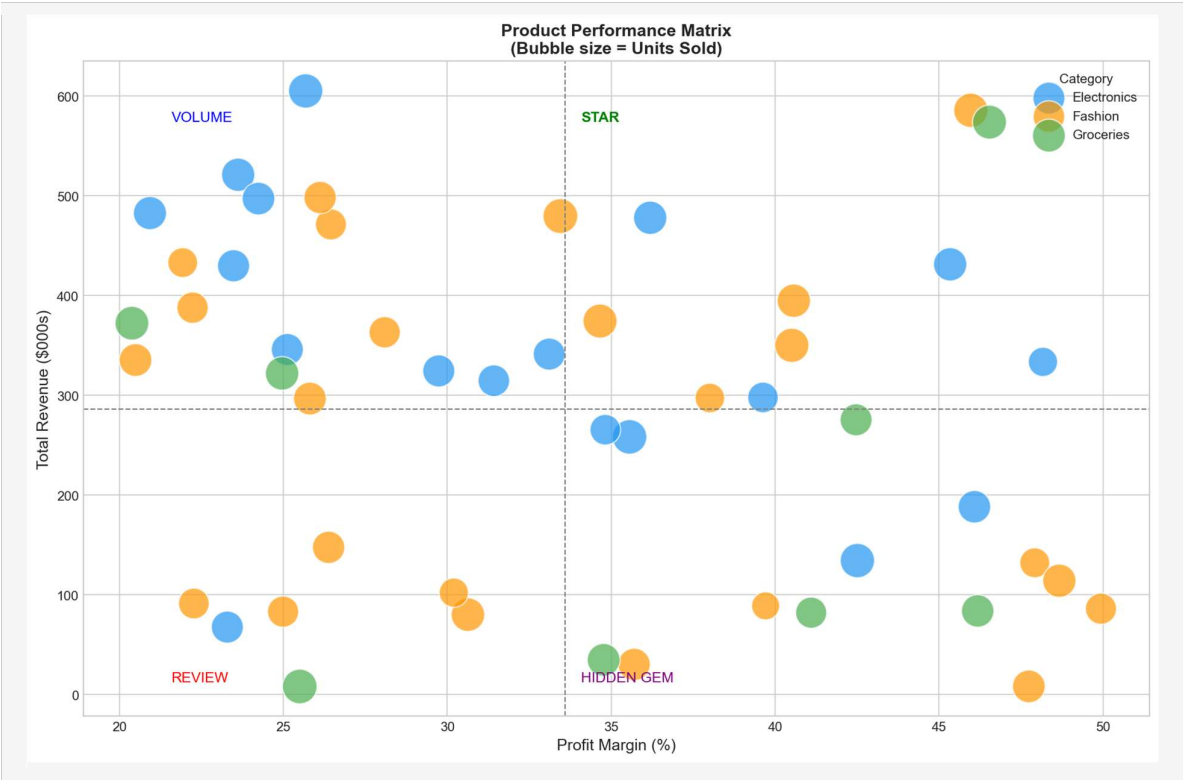
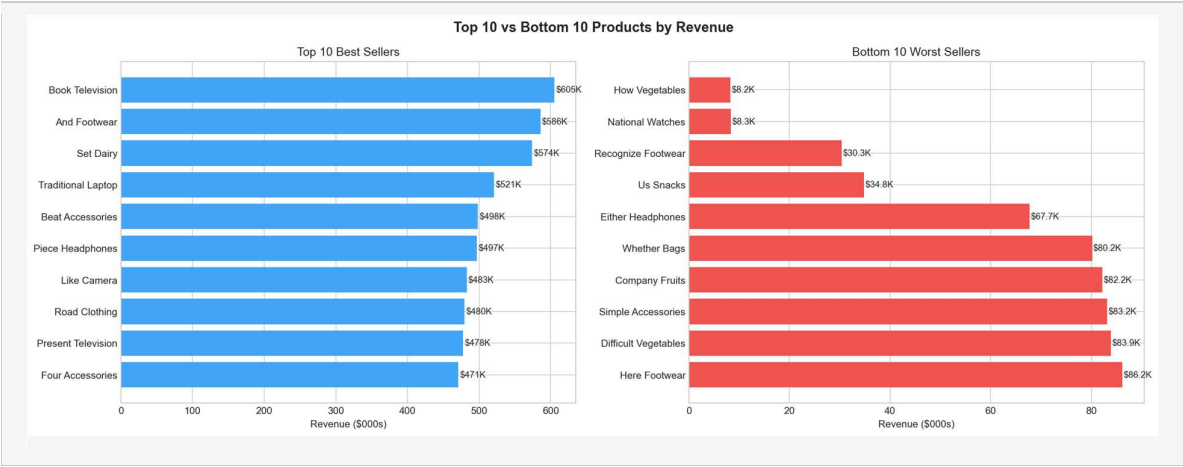
Category	Revenue	Share %	Transactions	Avg Margin %	Est. Profit
Electronics	\$6,316,917	44.2%	1,801	32.6%	\$2,056,907

Category	Revenue	Share %	Transactions	Avg Margin %	Est. Profit
Fashion	\$6,232,279	43.6%	2,365	33.7%	\$2,103,002
Groceries	\$1,752,705	12.3%	834	35.0%	\$612,665



Top 10 Best-Selling Products

Product	Category	Times Sold	Units	Revenue
Book Television	Electronics	115	337	\$605,028
And Footwear	Fashion	115	336	\$585,613
Set Dairy	Groceries	111	330	\$573,777
Traditional Laptop	Electronics	106	317	\$521,030
Beat Accessories	Fashion	103	299	\$498,224
Piece Headphones	Electronics	101	306	\$497,149
Like Camera	Electronics	102	314	\$482,608
Road Clothing	Fashion	111	346	\$479,596
Present Television	Electronics	101	317	\$477,913
Four Accessories	Fashion	98	275	\$471,228



5.4 Customer Insights

Demographics

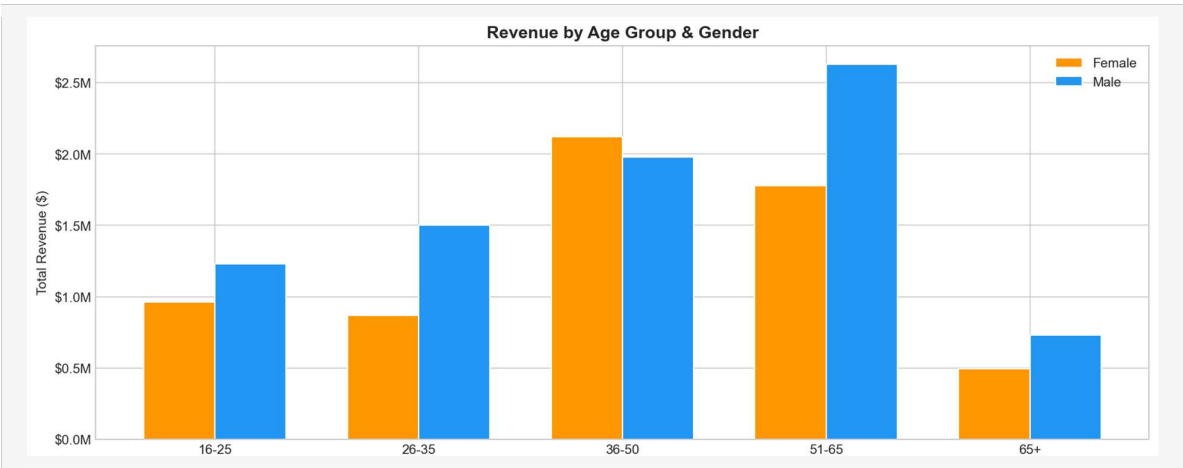
Gender	Customers	Avg Age	Youngest	Oldest
Male (M)	113	45	18	71
Female (F)	87	44	19	71

Age Group Distribution

Age Group	Customers
16–25 (Gen Z)	31
26–35 (Millennial)	33
36–50 (Gen X)	57
51–65 (Boomer)	60
65+ (Senior)	19

Top 5 Highest Spending Customers

Customer	Gender	Age	Orders	Total Spent	Avg Order
Dale Perry	M	36	32	\$117,085	\$3,658
Travis Peters	F	60	31	\$112,255	\$3,621
Juan Ramirez	M	46	34	\$111,867	\$3,290
Matthew Juarez	F	23	32	\$102,378	\$3,199
Anna Andrews	F	26	26	\$102,068	\$3,925



RFM Customer Segmentation

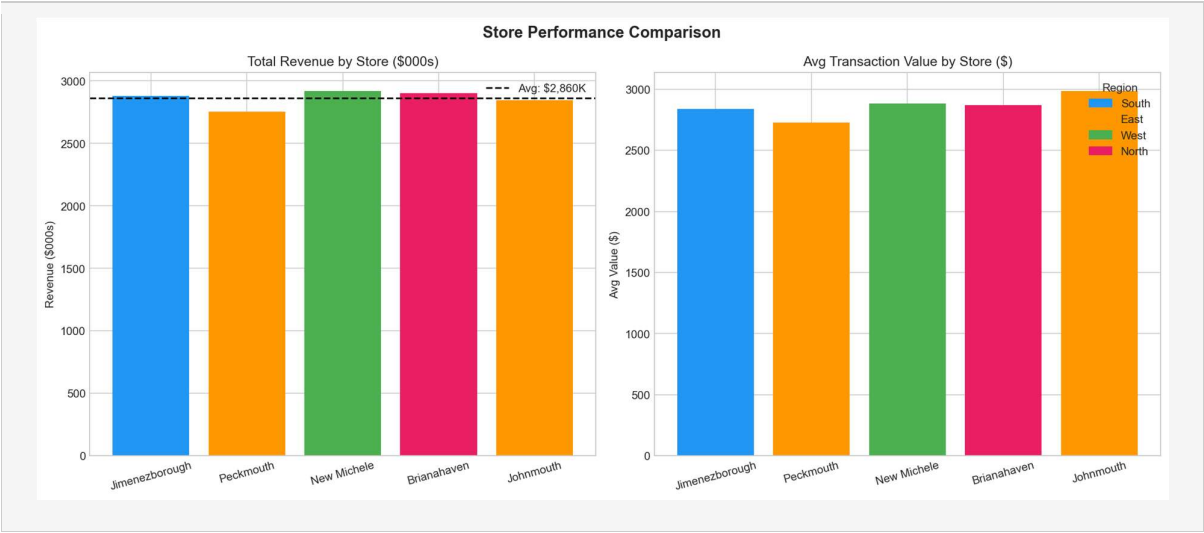
Customers were segmented using the RFM model (Recency, Frequency, Monetary) — an industry-standard marketing framework.

Segment	Customers	Avg Spend	Avg Orders
Champions	39	\$91,301	30.4
Loyal Customers	56	\$76,457	26.7
Potential Loyalists	57	\$66,654	23.9
At Risk	25	\$57,176	21.1
Lost Customers	23	\$53,513	18.7



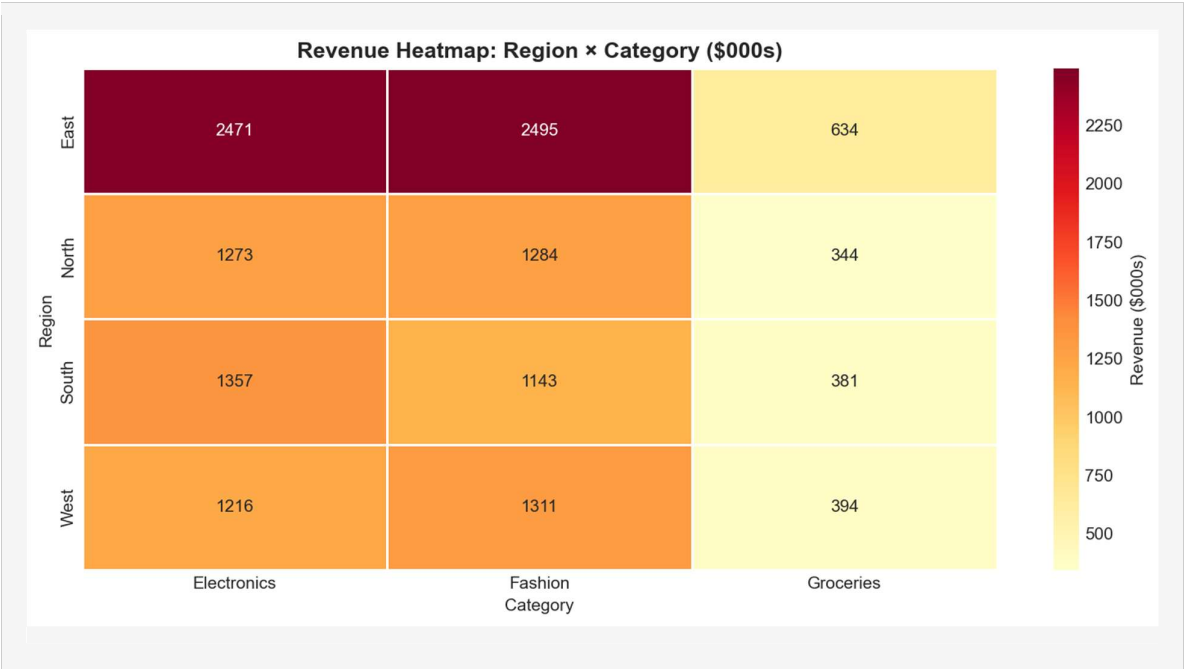
5.5 Store Performance

Store	Region	Revenue	Transactions	Avg Value	Performance
MegaMart New Michele	West	\$2,920,407	1,013	\$2,882	Above Avg
MegaMart Brianahaven	North	\$2,900,701	1,011	\$2,869	Above Avg
MegaMart Jimenezborough	South	\$2,880,114	1,014	\$2,840	On Target
MegaMart Johnmouth	East	\$2,847,047	953	\$2,987	Below Avg
MegaMart Peckmouth	East	\$2,753,632	1,009	\$2,729	Below Avg



Revenue by Region

Region	Stores	Revenue	Share %	Customers/Store
East	2	\$5,600,680	39.2%	100
West	1	\$2,920,407	20.4%	101
North	1	\$2,900,701	20.3%	100
South	1	\$2,880,114	20.1%	101



5.6 Payment & Discount Analysis

Payment Methods

Payment Method	Transactions	Revenue	Avg Value	Usage %
Cash	1,284	\$3,676,072	\$2,862	25.7%
Credit Card	1,281	\$3,673,747	\$2,867	25.6%
Mobile Money	1,276	\$3,558,433	\$2,788	25.5%
Bank Transfer	1,159	\$3,393,650	\$2,928	23.2%

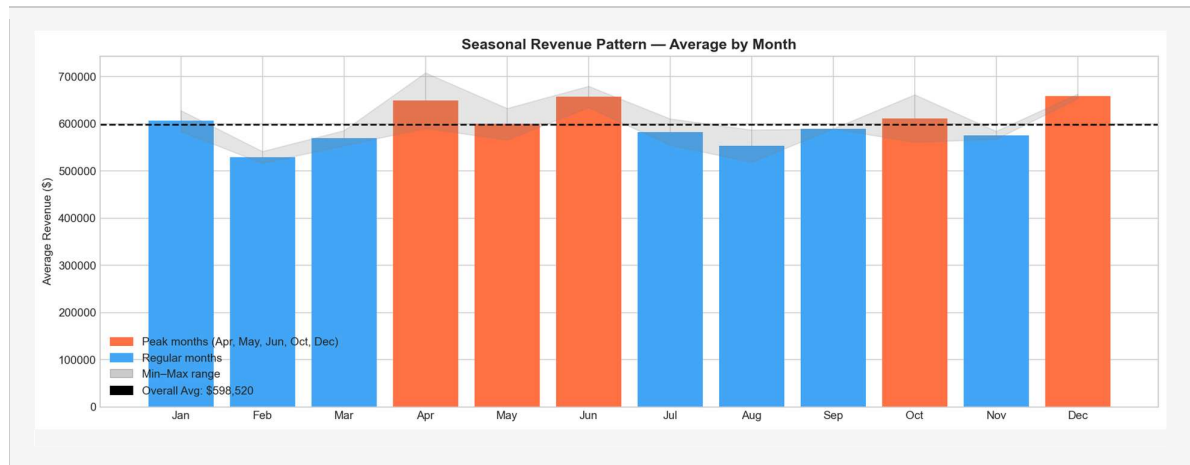
Discount Impact

Discount Tier	Transactions	Avg Qty	Avg Value	Total Revenue
No Discount (0%)	1,243	3	\$3,094	\$3,846,472
5% Discount	1,250	2	\$2,921	\$3,651,418
10% Discount	1,183	2	\$2,840	\$3,360,176
15% Discount	1,324	2	\$2,601	\$3,443,835



6. Trends, Patterns & Gap Analysis

6.1 Revenue Trends Identified



- ❖ Q2 (April–June) is the strongest season every year — April 2025 peaked at \$708,233
- ❖ December and October are secondary peaks, consistent across years
- ❖ February is consistently the weakest month in all years
- ❖ Overall revenue grew from \$2.18M (partial 2023) to \$7.03M (full year 2024)
- ❖ 2025 is on pace to exceed 2024 based on partial year performance

6.2 Customer Behaviour Patterns

- ❖ No customer made only 1 purchase — all 200 customers are repeat buyers (12–36 orders each)
- ❖ Gen X (36–50) and Boomers (51–65) contribute the most revenue by age group
- ❖ Male customers generate more total revenue (\$8.07M) vs Female (\$6.23M) reflecting gender ratio (113M vs 87F)
- ❖ Discounts do NOT drive higher spending — nodiscount transactions have the highest average value (\$3,094)
- ❖ Payment methods are evenly distributed across all 4 options — no dominant preference

6.3 Product Patterns

- ❖ Electronics and Fashion each contribute ~44% of revenue — Groceries is a distant third at 12.3%
- ❖ High-margin products (>45%) like "And Footwear" (45.97%), "Set Dairy" (46.54%), "Here Footwear" (49.94%) underperform in sales volume
- ❖ Top revenue products are driven by high unit prices, not necessarily high margins
- ❖ Some low-priced Groceries products like "How Vegetables" (\$8,233) and "National Watches" (\$8,327) represent near-zero revenue

6.4 Store & Region Gaps

- ❖ East region has 2 stores but generates 39.2% of revenue — highest demand-to-store ratio
- ❖ MegaMart Peckmouth consistently underperforms despite being in the high-revenue East region
- ❖ West, North, and South regions each have only 1 store contributing ~20% of revenue each
- ❖ MegaMart Jimenezborough had notably low performance in January 2024 (\$69,295) — potential operational issue

6.5 Gap Analysis Summary

Gap Identified	Impact	Recommendation
Groceries only 12.3% revenue share	High	Bundle promotions, volume deals, upselling at checkout
East region has highest demand with only 2 stores	High	Open 3rd East region store to capture unmet demand
MegaMart Peckmouth lowest revenue	Medium	Investigate staffing, location, product range, or operations
High-margin products have low visibility	Medium	Run targeted promotional campaigns for hidden gem products
Discounts not increasing basket value	Medium	Replace blanket discounts with loyalty rewards or bundles
At-risk customers (181-365 days inactive)	Medium	Launch re-engagement campaign with personalised offers

7. Business Decision Dashboard



7.1 Top 5 Business Recommendations

Priority	Recommendation	Evidence	Expected Impact
1 — HIGH	Expand East Region (3rd Store)	East has 2 stores generating 39% of revenue with highest customer density per store. All other regions are served by 1 store.	Potential \$2M+ additional annual revenue
2 — HIGH	Promote Hidden Gem Products	Products like "And Footwear" (45.97% margin), "Set Dairy" (46.54%), "Here Footwear" (49.94%) have high margins but low sales. Targeted campaigns could boost profit without heavy discounting.	Improve profit margin by 3–5%

Priority	Recommendation	Evidence	Expected Impact
3 — MEDIUM	Re-engage At-Risk Customers	Customers inactive for 181–365 days represent significant at-risk revenue. Each customer historically averages \$70K+ in lifetime spend.	Recover 10–20% of at-risk revenue
4 — MEDIUM	Plan Q2 Seasonal Campaigns	April–June peaks every year. April 2025 (\$708K), June 2025 (\$679K), and December 2024 (\$663K) were the top months. Early inventory and marketing planning is critical.	Increase peak month revenue by 10–15%
5 — MEDIUM	Investigate MegaMart Peckmouth	Consistently the lowest-revenue store despite being in the high-performing East region. Root cause analysis needed across staffing, product range, and local marketing.	Bring store to regional average (+\$150K revenue)

8. Project Files & Documentation

File	Location	Description
retail_sales_dataset.xlsx	data/	Original raw dataset from Kaggle (4 sheets)
customers_clean.csv	data/cleaned_data/	Cleaned customers table (200 rows, 9 columns)
products_clean.csv	data/cleaned_data/	Cleaned products table (50 rows, 7 columns)
stores_clean.csv	data/cleaned_data/	Cleaned stores table (5 rows, 4 columns)
transactions_clean.csv	data/cleaned_data/	Cleaned transactions table (5,000 rows, 9 columns)
retail_sales_preprocessing.ipynb	notebooks/	Data loading, EDA, cleaning, export
insights.ipynb	notebooks/	All visualizations and business dashboards
01_create_tables.sql	sql/	DDL: CREATE TABLE statements with all constraints
02_import_data.sql	sql/	BULK INSERT: Load all CSVs into SQL Server
03_queries_and_insights.sql	sql/	30+ analytical SQL queries across 9 sections
04_summary_and_trends.sql	sql/	Summaries, RFM, CLV, gap analysis, decisions

9. Conclusion

This project successfully completed all 8 phases of the data science internship task — from raw data collection and schema design through to business intelligence reporting.

A fully functional relational database was built in Microsoft SQL Server with proper referential integrity, constraints, and indexing. The dataset of 5,000 transactions across 200 customers, 50 products, and 5 store branches generated rich business insights through 40+ SQL queries organized into 16 analytical sections.

Key achievements of the project:

- ✓ Designed and implemented a clean star-schema relational database with 4 tables and full FK constraints
- ✓ Cleaned and enriched the dataset by deriving 3 new columns (FullName, Age, ProfitMargin, TotalAmount)
- ✓ Identified a clear Q2 seasonal peak pattern and year-on-year revenue growth trend
- ✓ Segmented 200 customers using the RFM model — identifying 39 Champion customers and 23 at-risk customers
- ✓ Classified all 50 products into performance tiers (Star, Volume, Hidden Gem, Review)
- ✓ Generated 5 prioritised business recommendations backed by data evidence
- ✓ Created 14 visualizations including an executive business intelligence dashboard

The most critical business insight is that the East region is significantly underserved — 2 stores generate 39% of revenue with the same customer density per store as single-store regions. Expanding East region capacity represents the highest-impact growth opportunity for the business.

End of Report

Data Science Internship | March 2026